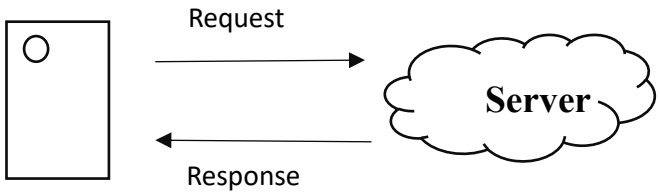
API Gateway

API : Application Programming Interface

API is a set of rules that allow two aaplications to communicate with each other.

Example : We use mobile app to check weather.



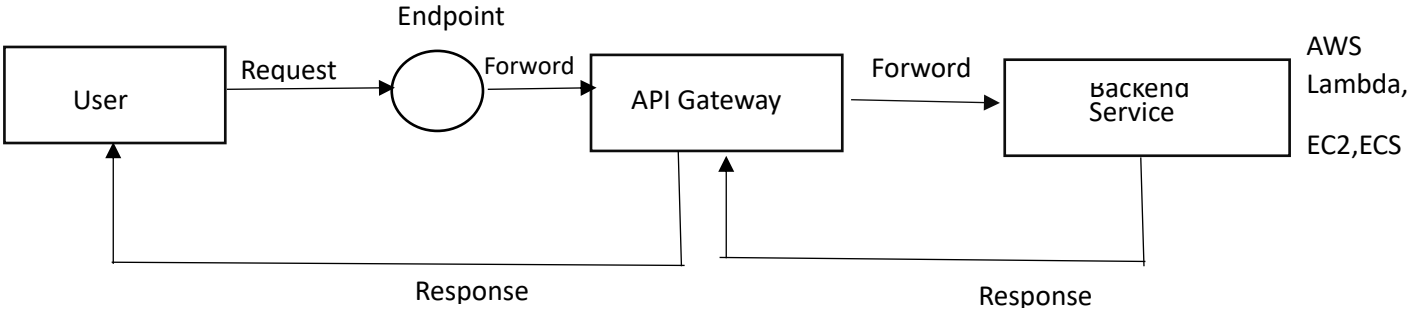
In Short : Messenger between client and server.

API Gateway : API Gateway is a fully managed service used to create,

publish,secure,monitor API's at any scale.

It act as a entry point for backend services like AWS

Lambda,EC2,ECS and other HTTP/HTTPS Services



Example : Security Gate at an office Building.

- All Visitors enter through one gate.
- The gate checks ID.
- Decides which office to send them to.
- Controls how many people can enter.
- Keeps records of entries.

Note : Client never go directly inside – they always go through the gate.

History :

Before API : Apps were standalone,could not share information.

API Came : Which helps apps to talk with each other.

Problem : When many apps talking with each other it becames confusing.

Solving : AWS Launched their API Gateway service in 2015.

Today : API Gateway are cloud native ,secure and supporting REST API ,

WEBSOCKET .

Why we use?

We use API Gateway to control, secure,and manage requests before they reach backend service.

Types of API Gateway

1) REST API

Handles HTTP / HTTPS Requests
Most common type

Methods :

1) GET : Fetch Data
2) POST : Create Data
3) PUT : Update Data
4) DELETE : Remove Data
5) PATCH : Partial Data

2) WEBSOCKET

Supports real-time two way communication .

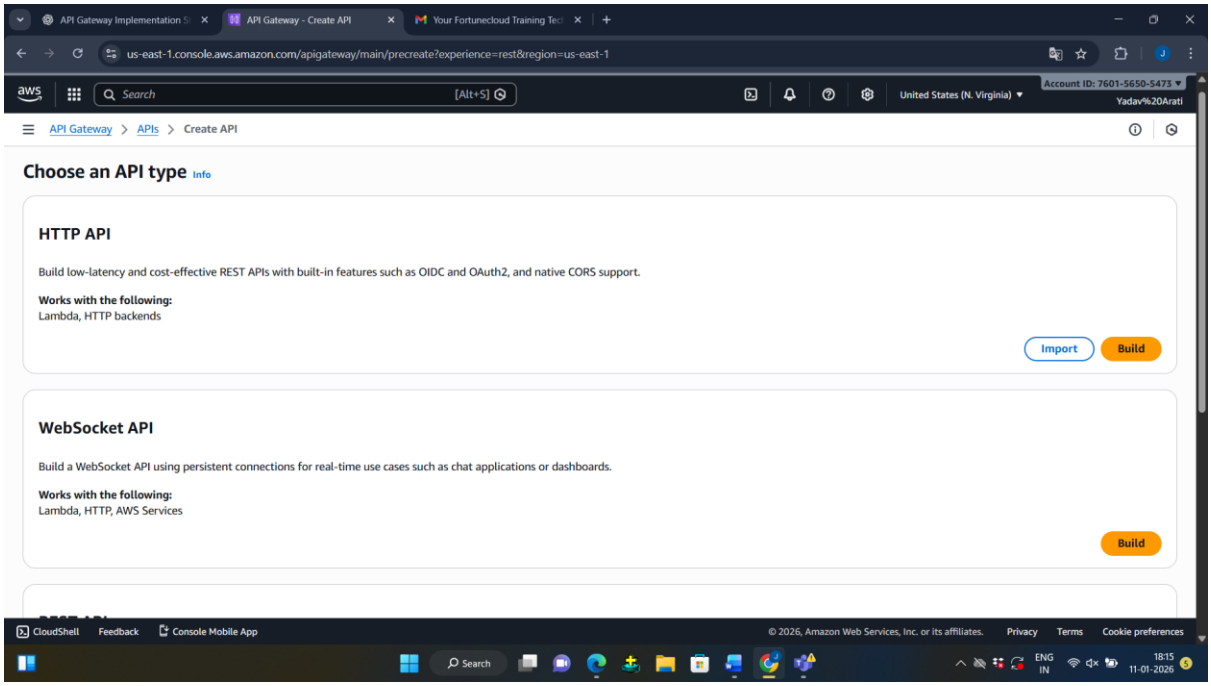
Example : Chat Application , Live Notification.

Implimentation :

Using HTTP :

1.Sign in to the AWS console

2. To create your first API, for HTTP API, choose Build.



3.For **API name**, enter **API**.

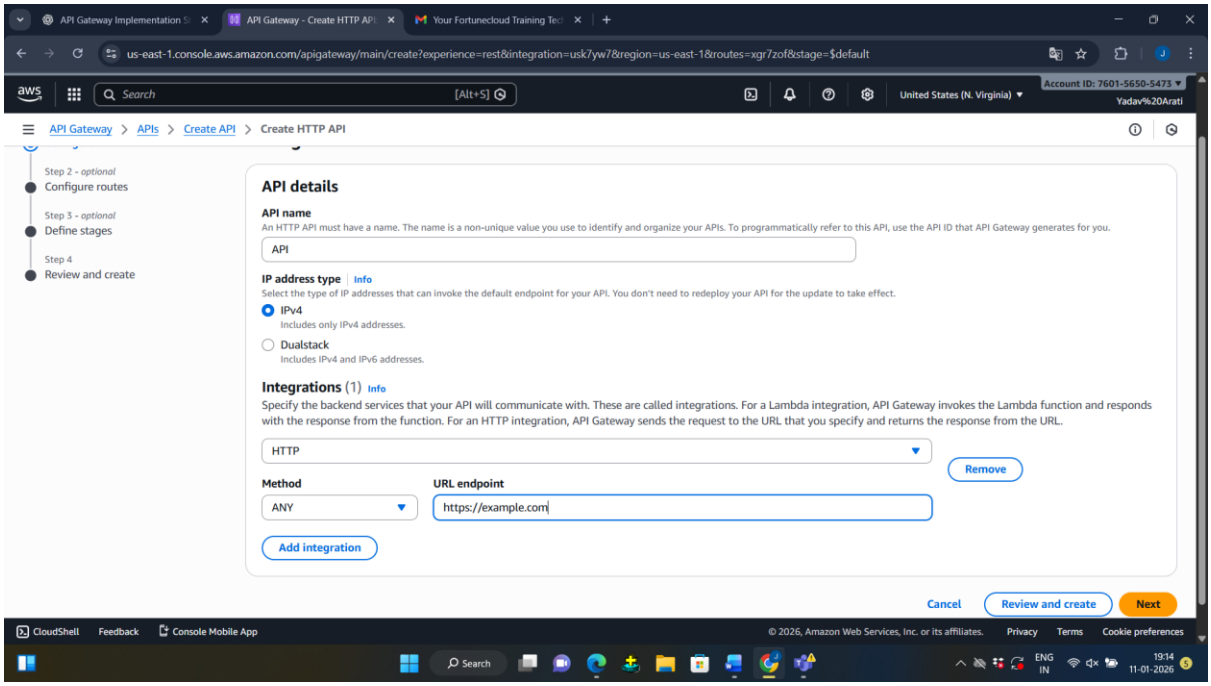
4.For **Integrations**, choose **Add integration**.

5.Choose **HTTPS**.

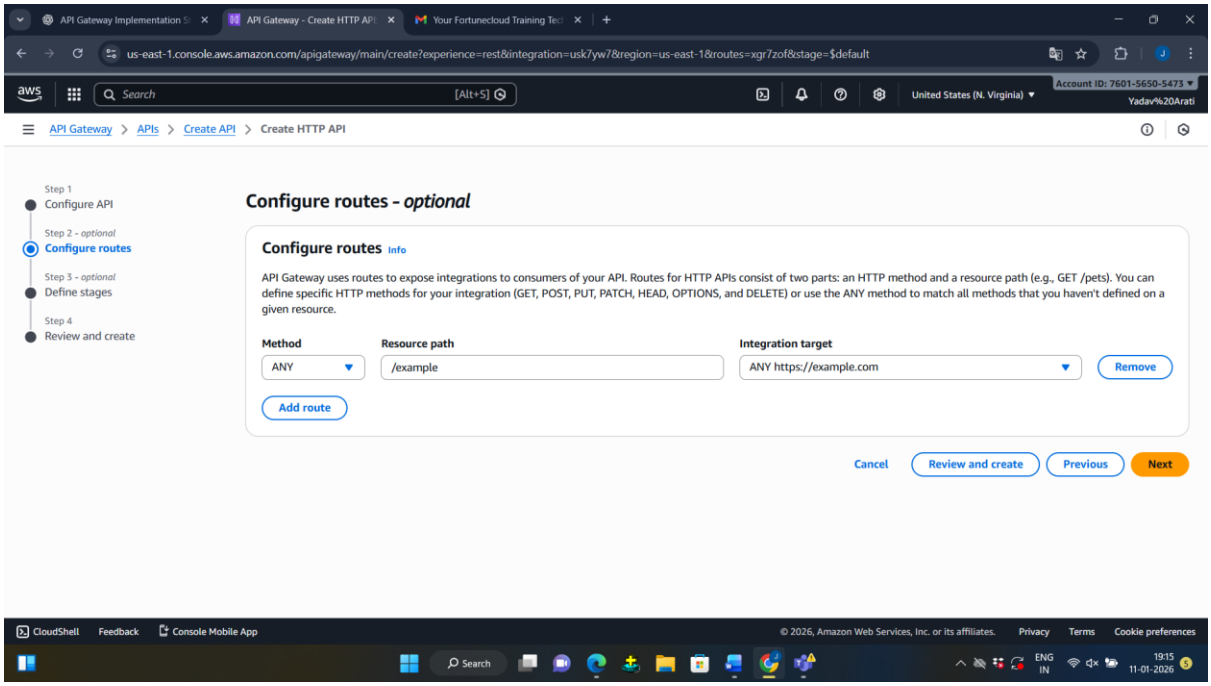
6.For **HTTP function**, enter **my-function**.

7. URL endpoint **: https://example.com**

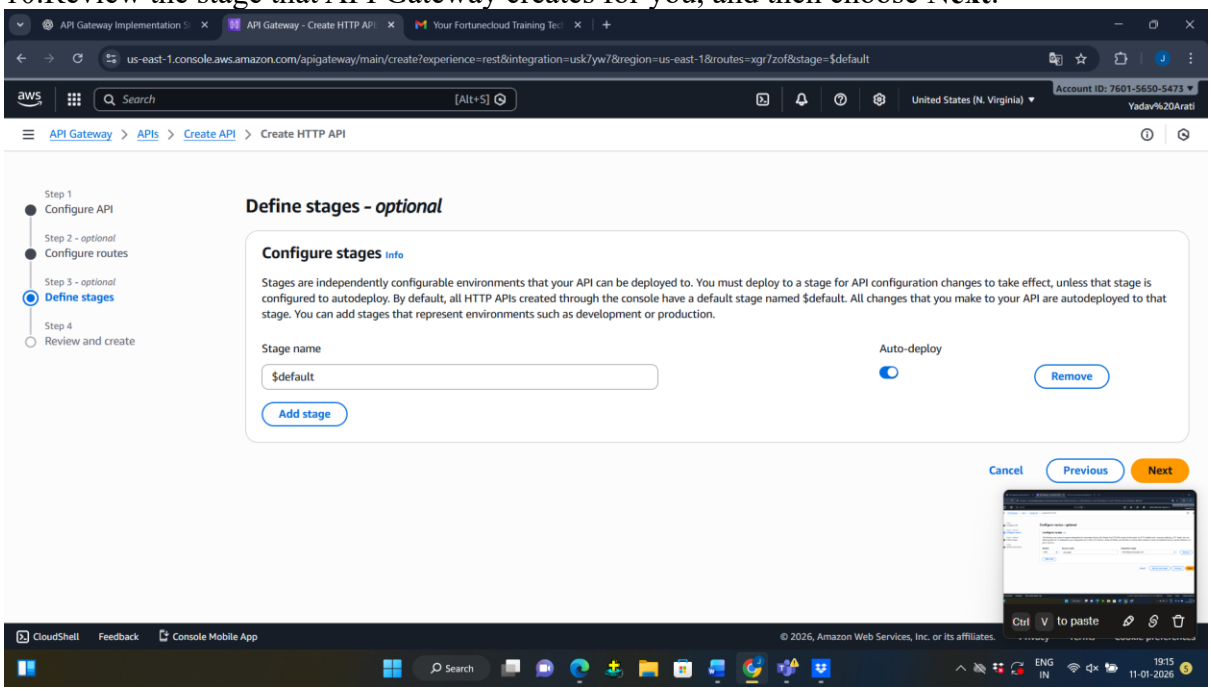
8.Choose **Next**.



9. Review the route that API Gateway creates for you, and then choose Next.



10.Review the stage that API Gateway creates for you, and then choose **Next**.



11.Choose **Create**.

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us-east-1.console.aws.amazon.com/apigateway/main/create?experience=rest&integration=usk7yw7&region=us-east-1&routes=xgr7zof&stage=$default

aws | Search [Alt+S] | United States (N. Virginia) | Account ID: 7601-5650-5473 | Yadav%20Arati

API Gateway > APIs > Create API > Create HTTP API

Step 1
Configure API

Step 2 - *optional*
Configure routes

Step 3 - *optional*
Define stages

Step 4
Review and create

Review and create

API name and integrations

Edit

API name
API

IP address type
IPv4

Integrations
- ANY https://example.com (HTTP)

Routes

Edit

Routes

- ANY /example → ANY https://example.com (HTTP)

Stages

Stages

- $default (Auto-deploy: enabled)

Ctrl V to paste

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us-east-1.console.aws.amazon.com/apigateway/main/develop/routes?api=e463q16f7c&experience=rest&integration=usk7yw7&region=us-east-1&routes=xgr7zof&stage=$default

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API Gateway > APIs > Routes - API (e463q16f7c)

API Gateway

APIs
Custom domain names
Domain name access associations
VPC links

API: API(e463q16f7c)

- Develop
  - Routes
  - Authorization
  - Integrations
  - CORS
  - Reimport
  - Export
- Deploy
  - Stages
- Monitor
  - Metrics

Successfully created API API (e463q16f7c).

Routes

Stage: - | Deploy

Routes for API

Create

Search

▼ /example
ANY

Choose a route.

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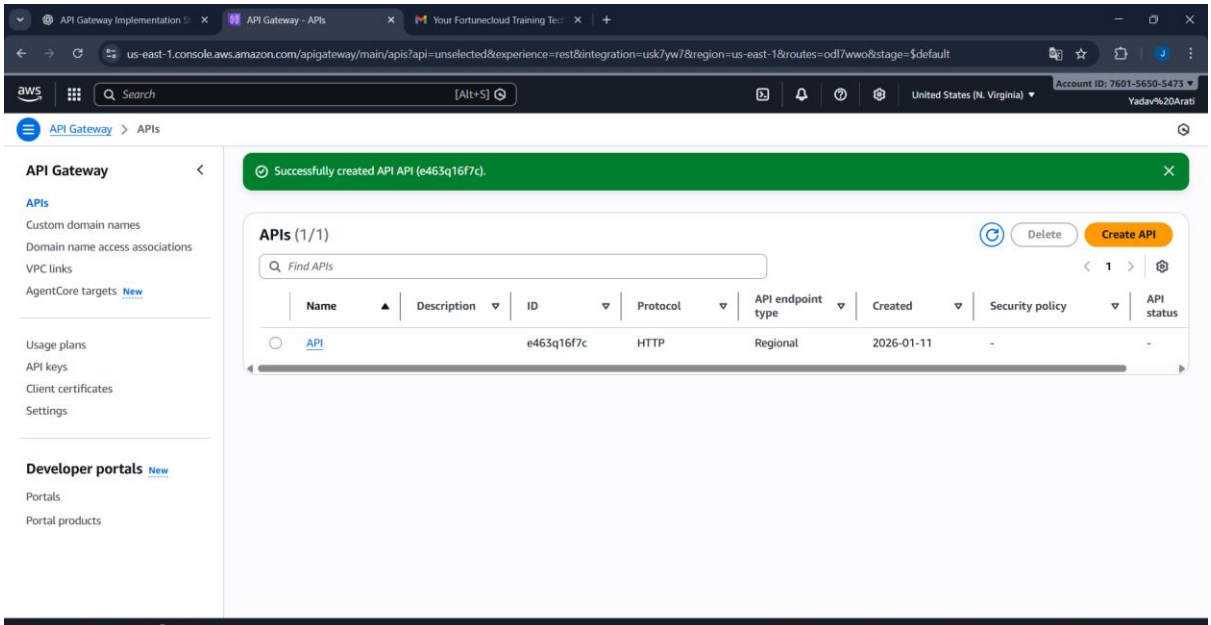
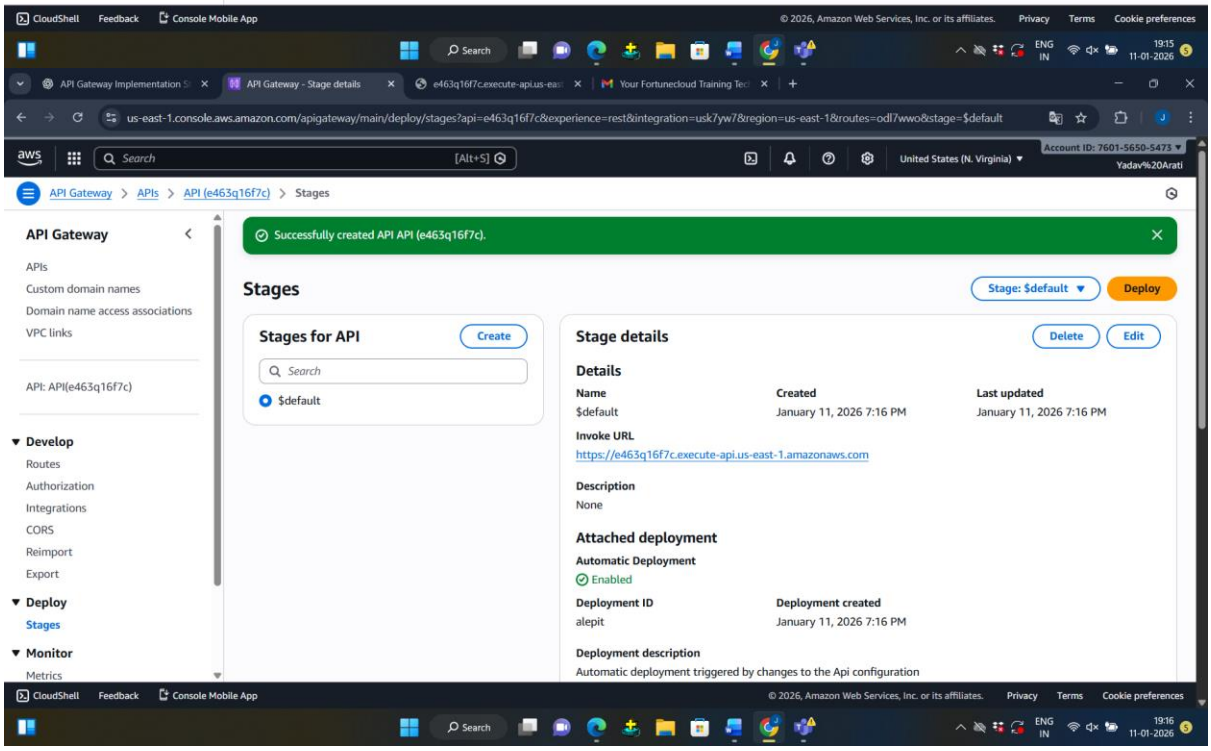
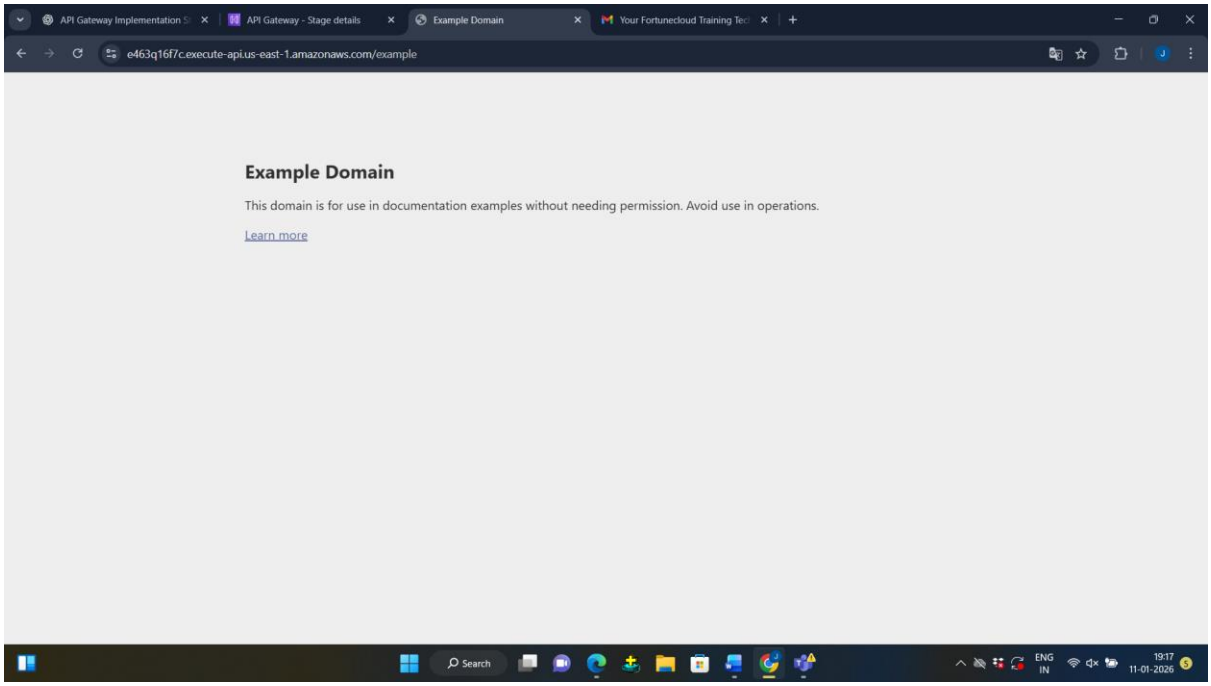




12. Now you've created an HTTP API

Test your API

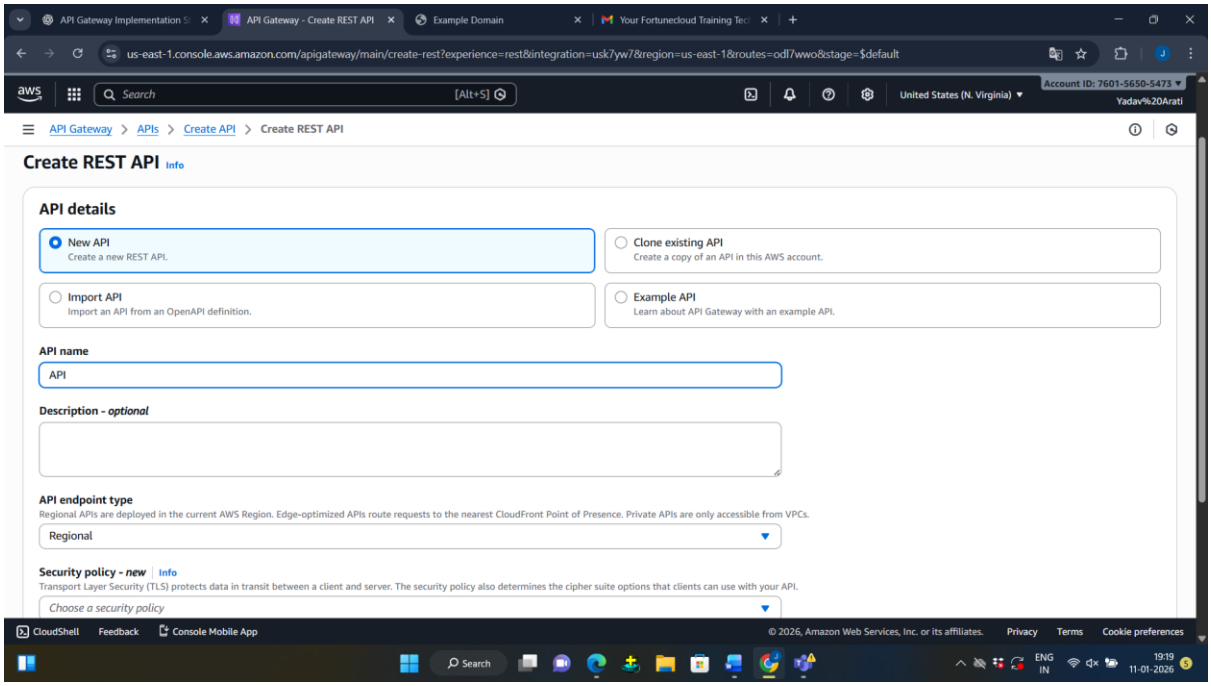
Next, you test your API to make sure that it's working. For simplicity, use a web browser to invoke your API.



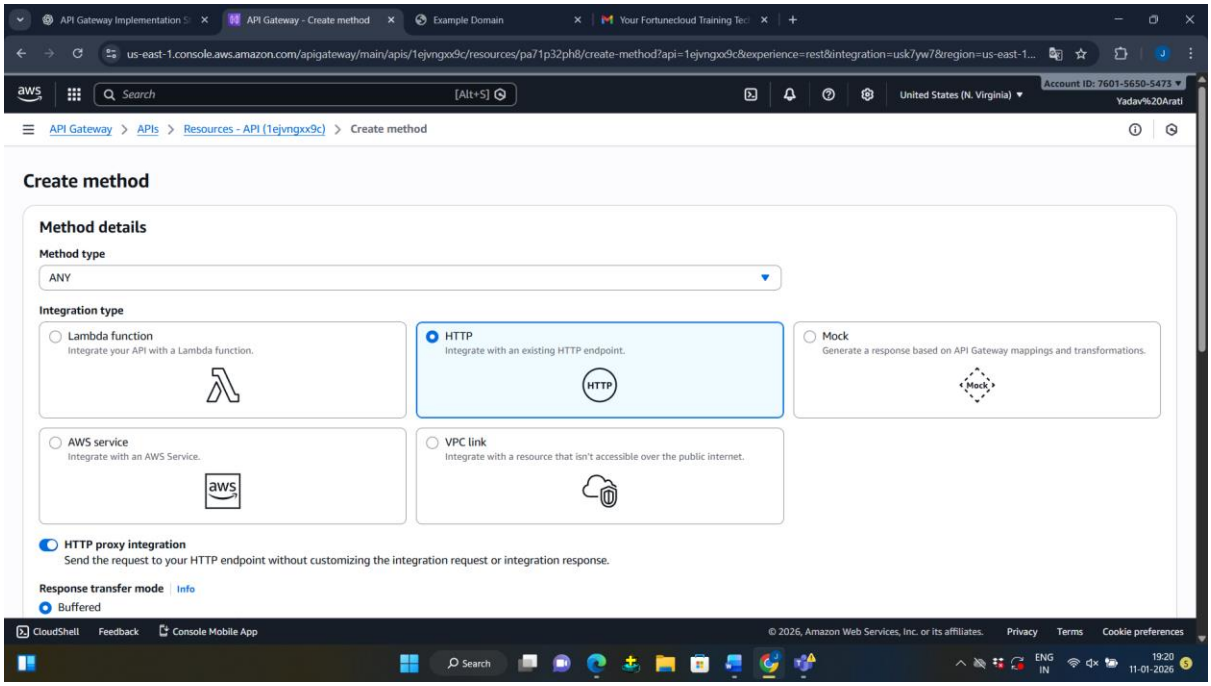
Using REST API :

1. **Login to AWS Console**

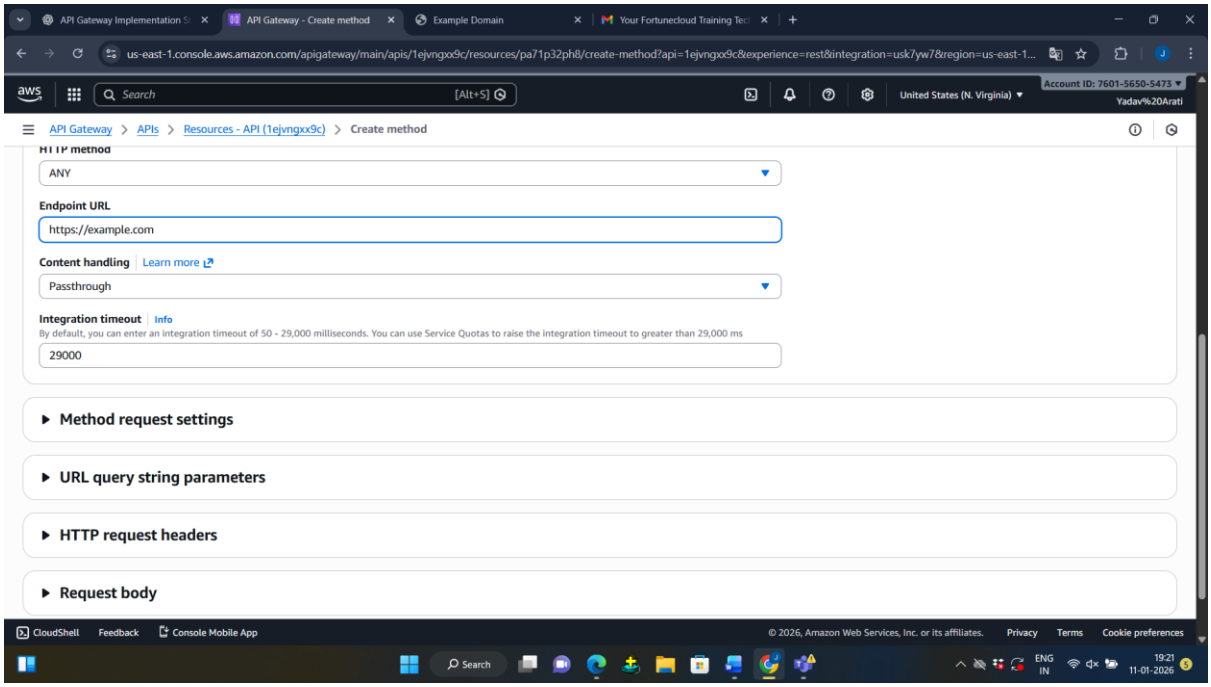
2. **Create a REST API**



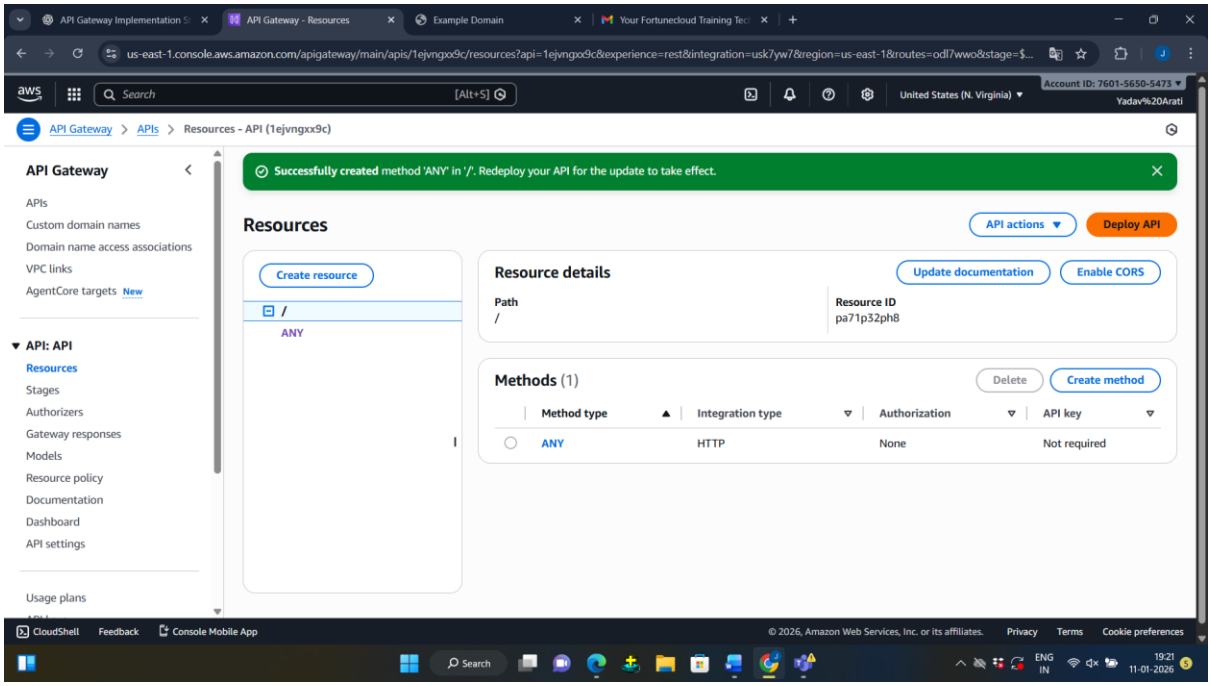
3. **Create a Method (GET, POST, etc.)**



4. Connect to Backend



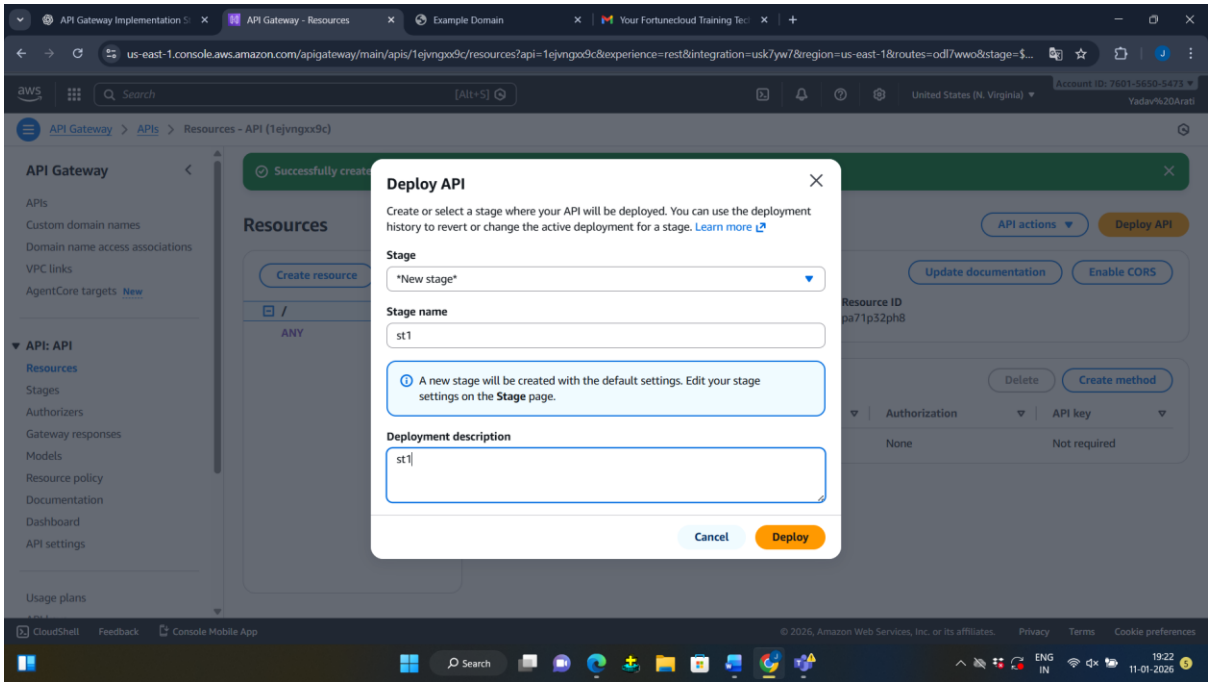
5. Deploy the API

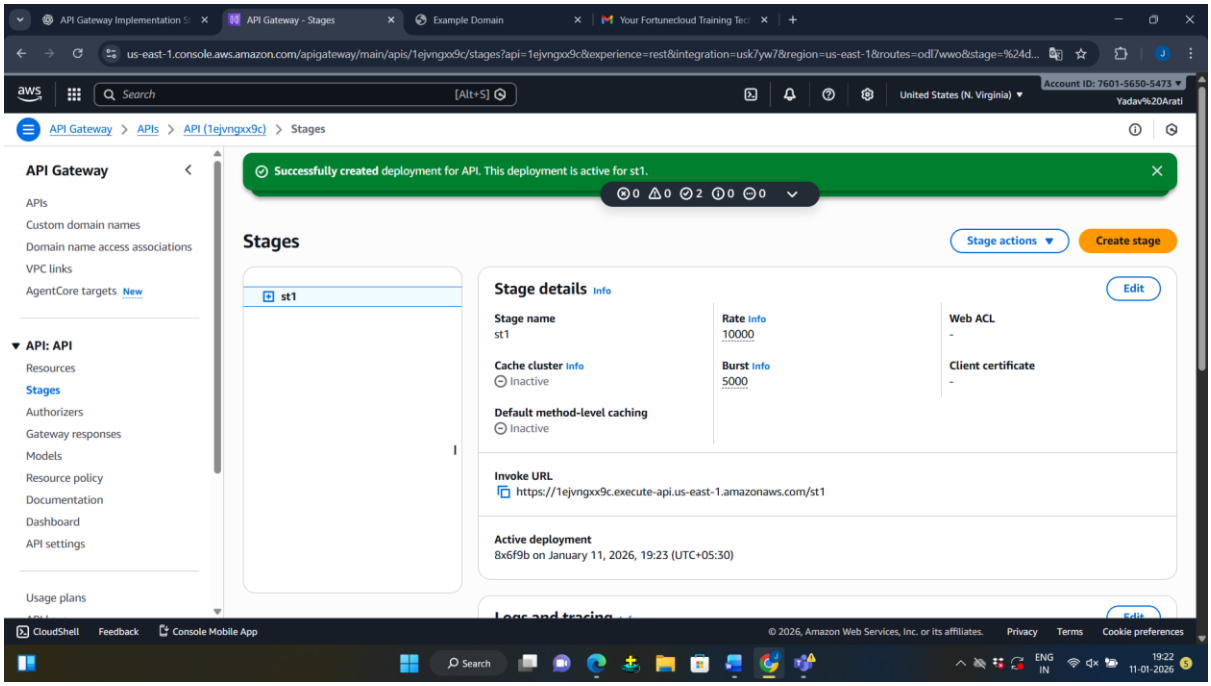


6. Create new stage:

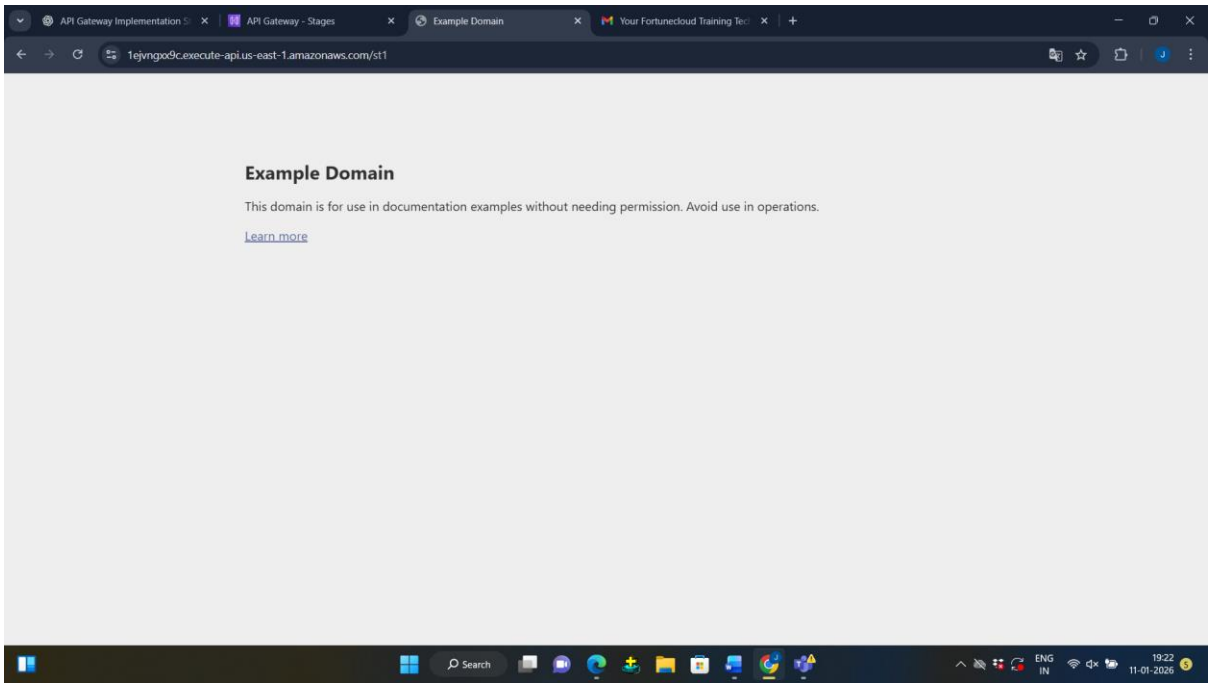
- **Stage name: st1**

7. Click Deploy





8. Get Your Endpoint

Test your API

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